

1. Please answer the following questions:

- (a) Illustrate the operation of RADIX-SORT on the following list of English words: COW, DOG, SEA, RUG, ROW, MOB, BOX, TAB, BAR, EAR, TAR, DIG, BIG, TEA, NOW, FOX.
- (b) Use induction to prove that radix sort works. Where does your proof need the assumption that the intermediate sort is stable?

(a)

	2分	2分	2分
COW	SEA	TAB	BAR
DOG	TEA	BAR	BIG
SEA	MOB	EAR	BOX
RUG	TAB	TAR	COW
ROW	DOG	SEA	DIG
MOB	RUG	TEA	DOG
BOX	DIG	DIG	EAR
TAB	BIG	BIG	FOX
BAR	BAR	MOB	MOB
EAR	EAR	DOG	NOW
TAR	TAR	COW	ROW
DIG	COW	ROW	RUG
BIG	ROW	NOW	SEA
TEA	NOW	BOX	TAB
NOW	BOX	FOX	TAR
FOX	FOX	RUG	TEA

每條錯 16 $\times \frac{3}{4}$ 個以上扣 2 分, 其它扣 1 分

(b) 在 induction step 時, 考慮到第 i 個位元, 如果第 i 個位元大小相同, 需依照前 $i-1$ 個位元的大小做排序, 所以要用 stable sort

4 分

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- (b) Use induction to prove that radix sort works. Where does your proof need the assumption that the intermediate sort is stable?



(b)

在 Inductive Step 排序 $k+1$ 位的元素時由歸納假設得知，因前 $1 \sim k$ 位元素已排好，且排序穩定，所以遇到相同元素時，其相對位置不變。

命題 → 對 d 位數字，Radix Sort 執行 d 輪穩定排序後結果正確。

Base Case: $d=1$

只有一位數，直接對該位做一輪穩定排序 → 顯然正確。

Inductive Step:

假設前 k 輪排序後，所有元素已依照第 k 低位 ~ 第 1 低位正確排序。

— 第 $k+1$ 位不同的元素：依照第 $k+1$ 位來排好

— 第 $k+1$ 位相同的元素：因排序穩定，相對順序不變，保留 $1 \sim k$ 位已排好的順序

所以 $k+1$ 輪後，元素會依照 $k+1$ 位到第 1 位的順序正確排序

由歸納法得證，經過 d 輪後排序正確。

1. Please answer the following questions:
 - (a) Illustrate the operation of COUNTING-SORT on the array $A = \langle 7, 3, 6, 0, 1, 2, 7, 6, 4, 8, 7 \rangle$.
 - (b) Illustrate the decision tree for insertion sort operating on four elements.

(a) Illustrate the operation of COUNTING-SORT on the array $A = \langle 7, 3, 6, 0, 1, 2, 7, 6, 4, 8, 7 \rangle$.

(b) Illustrate the decision tree for insertion sort operating on four elements.

(a) ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~

$A < 7, 3, 6, 0, 1, 2, 7, 6, 4, 8, 7 >$

$C < \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1, & 1, & 1, & 1, & 1, & 0, & 2, & 3, & 1 \end{matrix} >$

$\hookrightarrow C < \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1, & 2, & 3, & 4, & 5, & 5, & 7, & 10, & 11 \end{matrix} >$

$B < \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ & & & & & & & & & 7 & \end{matrix} >$
 $C < \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 & 5 & 5 & 7 & 9 & 11 \end{matrix} >$
 $\downarrow$
 $B < \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ & & & & & & & & & 7 & 8 \end{matrix} >$
 $C < \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 & 5 & 5 & 7 & 9 & 10 \end{matrix} >$
 $\downarrow$
 $B < \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ & & & & 4 & & & & & 7 & 8 \end{matrix} >$
 $C < \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 & 5 & 5 & 7 & 9 & 10 \end{matrix} >$
 $\downarrow$
 $B < \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 \\ & & & & 4 & & 6 & & & 7 & 8 \end{matrix} >$
 $C < \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 & 4 & 5 & 9 & 10 \end{matrix} >$
 $\downarrow$

	1	2	3	4	5	6	7	8	9	10	11	
B <					4		6		7	7	8	>
	0	1	2	3	4	5	6	7	8			
C <	1	2	3	4	4	5	6	9	10			
								8				
					↓							
	1	2	3	4	5	6	7	8	9	10	11	
B <			2		4		6		7	7	8	>
	0	1	2	3	4	5	6	7	8			
C <	1	2	3	4	4	5	6	8	10			
			2									
					↓							
	1	2	3	4	5	6	7	8	9	10	11	
B <		1	2		4		6		7	7	8	>
	0	1	2	3	4	5	6	7	8			
C <	1	2	2	4	4	5	6	8	10			
		1										
					↓							
	1	2	3	4	5	6	7	8	9	10	11	
B <	0	1	2		4		6		7	7	8	>
	0	1	2	3	4	5	6	7	8			
C <	1	1	2	4	4	5	6	8	10			
	0											
					↓							

	1	2	3	4	5	6	7	8	9	10	11
B <	0	1	2		4	6	6		7	7	8 >

原陣列 A 由右至左掃

	0	1	2	3	4	5	6	7	8	
C <	0	1	2	4	4	5	6	8	10 >	

以 A[idx] 的值當作 C 的 idx, 並將其結果作 B 的 idx, 放入 B 中。

↓

	1	2	3	4	5	6	7	8	9	10	11
B <	0	1	2	3	4	6	6		7	7	8 >

	0	1	2	3	4	5	6	7	8	
C <	0	1	2	4	4	5	5	8	10 >	

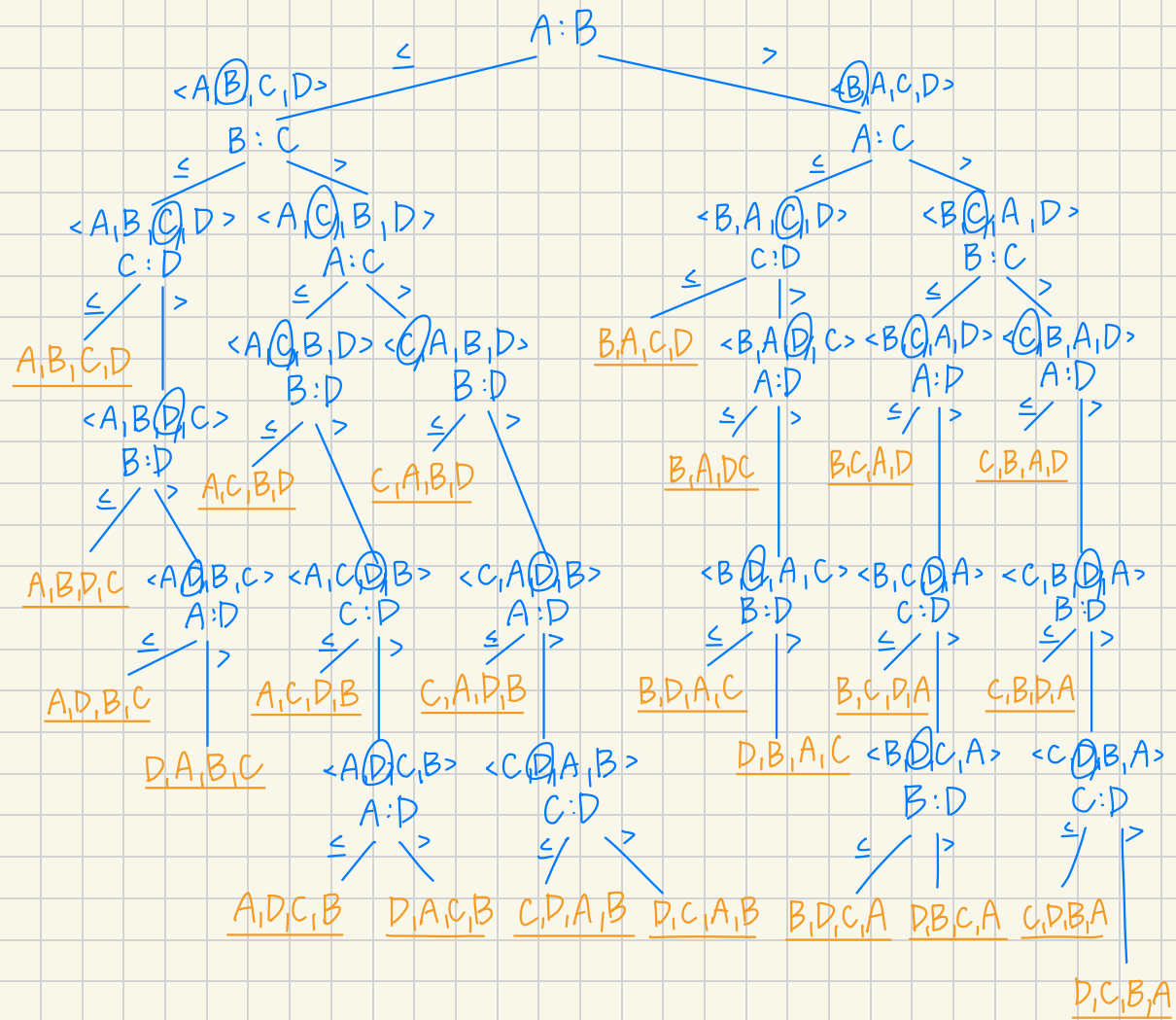
↓

	1	2	3	4	5	6	7	8	9	10	11
B <	0	1	2	3	4	6	6	7	7	7	8 >

	0	1	2	3	4	5	6	7	8	
C <	0	1	2	3	4	5	5	8	10 >	

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(b) $O \rightarrow$ 上一步比較insert的element
 $\langle A, B, C, D \rangle$



1. Please answer the following questions:

- (a) Illustrate the operation of BUCKET-SORT on the array $A = (.24, .17, .16, .64, .39, .56, .89, .53, .71, .42)$.
- (b) Describe an algorithm that, given n integers in the range 0 to k , preprocesses its input and then answers any query about how many of the n integers fall into a range $[a, b]$ in $O(1)$ time. Your algorithm should use $\Theta(k + n)$ preprocessing time.

(a)

A	B		B	
.24	0	/	0	/
.17	1	○ → .17 → .16	1	○ → .16 → .17
.16	2	/	2	/
.64	3	○ → .39	3	○ → .39
.39	4	○ → .42	4	○ → .42
.56	5	○ → .56 → .53	5	○ → .53 → .56
.89	6	○ → .64	6	○ → .64
.53	7	○ → .71	7	○ → .71
.71	8	○ → .89	8	○ → .89
.42	9	/	9	/

sort B
→

0	/
1	○ → .16 → .17
2	/
3	○ → .39
4	○ → .42
5	○ → .53 → .56
6	○ → .64
7	○ → .71
8	○ → .89
9	/

sort B →

→ $\langle .16, .17, .39, .42, .53, .56, .64, .71, .89 \rangle$

(b)

* A: 原始陣列

RangeQuery(A, n, k, a, b)

前處理 $\Theta(k+n)$

- create array $C[k]$ // 計數 $0 \sim k$ 出現次數, 算前綴和
- for $i \leftarrow 0 \sim k$
 - $C[i] \leftarrow 0$
- for $i \leftarrow 0 \sim n$
 - $C[A[i]] \leftarrow C[A[i]] + 1$
- for $i \leftarrow 1 \sim k$
 - $C[i] \leftarrow C[i] + C[i-1]$

將計數數器清0, $\Theta(k)$

算一個元素出現過幾次, $\Theta(n)$

維護C的前綴和, $\Theta(k)$

查詢 $O(1)$

- if $a > 0$
 - return $C[b] - C[a-1]$
- else if $a = 0$
 - return $C[b]$